

Full Answer Key – Activities 1–6

Answers for every Basic and Challenge exercise. Bonus activities are open-ended and don't have single correct answers — see grading notes.

Activity 1: What is an Algorithm?

BASIC — Write 5 Sandwich Steps

Open-ended. Grade against the Self-Check criteria: at least 5 steps, steps in a workable order, steps are specific (e.g., "open the bread bag," not "get bread").

CHALLENGE — Test Your Algorithm on Someone

Open-ended. The goal is that the tester follows the steps literally and the kid identifies what was missing or ambiguous. Any honest gap they spot (didn't say to open the jar, didn't mention a knife) counts as success.

BONUS Cut & Sort correct order:

1. Open the bread bag
2. Take two slices of bread
3. Get a butter knife
4. Spread peanut butter
5. Close the sandwich

(Accept swapping steps 2 and 3.)

Activity 2: Sequencing Puzzle – Morning Routine

BASIC — Number the 8 morning steps

- Wake up
- Wash face
- Brush teeth
- Get dressed
- Eat breakfast
- Pack school bag
- Put on shoes
- Go to school

(Accept reasonable variations: wash face / brush teeth may swap; eat breakfast may come before get dressed. "Wake up" must be first; "Go to school" must be last.)

CHALLENGE — Spot the Broken Sequence

- **Algorithm A (Making Cereal):** Step 3 "Eat the cereal" is in the wrong place — it should be LAST, after getting the spoon.
- **Algorithm B (Soccer Practice):** Step 1 "Put on your soccer cleats" is in the wrong place — cleats go on AFTER socks, so it belongs after step 2.

BONUS Brushing Your Teeth correct order:

1. Pick up your toothbrush
2. Wet the toothbrush under the tap
3. Put toothpaste on the toothbrush
4. Brush your teeth for 2 minutes
5. Spit out the toothpaste and rinse your mouth
6. Rinse the toothbrush and put it back

(Accept wetting the brush before or after applying toothpaste.)

Activity 3: Spot the Loop

BASIC — Find the Repeat

- **Song:** "If you're happy and you know it, clap your hands (clap clap)" repeats — 3 times.
- **P.E. Warm-Up:** "Do 10 jumping jacks" repeats 3 times (with "rest 5 seconds" repeating twice in between).
- **Pancakes:** The whole pour → wait → flip → cook → remove cycle repeats once per pancake — 5 times for 5 pancakes.
- **Brushing Teeth:** Brushing repeats twice per day (morning + night) — 14 times in a week.

CHALLENGE — Loop Detective

- **Scenario A:** Repeat "call the next name and hear 'Here!'" 25 times.
- **Scenario B:** Repeat "set one place (plate, fork, knife, spoon, napkin)" 4 times.

BONUS Spot the Loop in a Pattern

1. **Pattern A:** Triangle, circle, square repeats. Next 3 shapes: triangle, circle, square.
2. **Pattern B:** The loop is 4 moves: Step Left → Clap → Step Right → Clap.
3. **Pattern C:** Trash / Recycling alternates. Loop length: 2 days.

Activity 4: Write Your Own Loop

BASIC — Loop It!

- Scenario 1: REPEAT "do a push-up" 20 TIMES
- Scenario 2: REPEAT "water one plant" 6 TIMES
- Scenario 3: REPEAT "high-five one friend" 8 TIMES
- Scenario 4: REPEAT "put one sticker on a card" 12 TIMES

CHALLENGE

- **Part A A1:** REPEAT "pour water into glass" UNTIL glass is full
- **Part A A2:** REPEAT "practice free throw" UNTIL 10 in a row are made
- **Part B:** REPEAT 4 TIMES { REPEAT "hand quiz to next student" 5 TIMES }

BONUS Grading Note: Robot cleaning loop. Accept any valid loop using the robot's allowed actions.

Activity 5: Yes/No Conditionals

BASIC — Read the Flowchart

- **Flowchart 1** (rainy Monday): Bring an umbrella
- **Flowchart 2** (homework half-done): Stay and finish it (half-done ≠ finished)
- **Flowchart 3** (25°C outside): Wear pants (25 is NOT above 30)

CHALLENGE — Build Your Own Conditional

- **Part A A1:** IF you put in \$2 THEN give snack OTHERWISE return money
- **Part A A2:** IF it's 8:00 AM THEN go inside OTHERWISE keep playing
- **Part A A3:** IF you have a library card THEN borrow the book OTHERWISE sign up at the desk
- **Part B:** Diamond "Is temperature below 15°C?" → Yes: "Wear a jacket" / No: "Leave it at home"

BONUS Grading Note: Real-world conditional hunt. Accept any 4 valid IF-THEN-OTHERWISE conditionals from real life.

Activity 6: Nested Conditionals

BASIC — Follow the Decision Tree

- **Tree 1 Q1** (starving, has pizza): Eat pizza
- **Tree 1 Q2** (not hungry): Wait until dinner
- **Tree 2 Q1** (sunny Saturday): Go to the park
- **Tree 2 Q2** (Tuesday): Go to school

CHALLENGE — Build a Nested Conditional

- **Part A A1:** IF it's a school day THEN { IF raining THEN wait in cafeteria OTHERWISE go outside } OTHERWISE stay home
- **Part A A2:** IF book is overdue THEN pay a fine OTHERWISE { IF book is damaged THEN pay damage fee OTHERWISE no charge }
- **Part B Bug:** Inner OTHERWISE says "walk the dog outside" — but the outer OTHERWISE already means the weather isn't good. Contradiction.
Fix: change the inner OTHERWISE to "give the dog a treat."

BONUS Grading Note: Open-ended decision tree. Accept any 2-question tree that produces 3 distinct outcomes.